

PES UNIVERSITY

(Established under Karnataka Act No. 16 of 2013)

100-ft Ring Road, Banashankari Stage III, Bengaluru — 560085, Karnataka, India

A Project Report

On

**Zero-Variance Fraction (ZVF):
A Diagnostic Benchmark for GRPO
Post-Training of Language Models**

Measurement, Diagnostics, and Reproducibility Tooling for RLVR

Submitted in partial fulfilment of the requirements for
Project Phase-1 of the degree of

Master of Technology

in

Data Science and Machine Learning

Submitted by

Arvind C R

SRN: PES2PGE24DS140

Under the guidance of

Ramesh Prakash Guledgudd

Project Guide

Course: UE20CS971 — Project Phase-1 | May 2026 — July 2026

FACULTY OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

2026

PES UNIVERSITY, BENGALURU
FACULTY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the Project Phase-1 report entitled “**Zero-Variance Fraction (ZVF): A Diagnostic Benchmark for GRPO Post-Training of Language Models**” is a bona fide record of the work carried out by **Arvind C R** (SRN **PES2PGE24DS140**), a student of the Master of Technology programme in Data Science and Machine Learning, PES University, in partial fulfilment of the requirements for the course **UE20CS971 — Project Phase-1**, during the academic year 2025–2026. This is an *individual* project; the candidate has worked on a unique project title and has independently carried out the work reported herein.

The work presented in this report has been carried out under my supervision and guidance. To the best of my knowledge, the contents of this report, in full or in parts, have not been submitted to any other institute or university for the award of any degree or diploma.

Ramesh Prakash Guledgudd

Project Guide
Dept. of Computer Science and
Engineering
PES University

Chairperson

Dept. of Computer Science and
Engineering
PES University

External Viva

Name and Signature of Examiner 1

Name and Signature of Examiner 2

DECLARATION

I, **Arvind C R** (SRN **PES2PGE24DS140**), hereby declare that the Project Phase-1 report entitled “**Zero-Variance Fraction (ZVF): A Diagnostic Benchmark for GRPO Post-Training of Language Models**”, submitted in partial fulfilment of the requirements for the course **UE20CS971 — Project Phase-1** of the Master of Technology programme in Data Science and Machine Learning, is an original piece of work carried out by me under the guidance of **Ramesh Prakash Guledgudd**.

I further declare that the matter embodied in this report has not been submitted, in full or in parts, to any other institute or university for the award of any degree or diploma. All sources of information and prior work drawn upon have been duly acknowledged and cited. The empirical findings reported herein — including the negative and null results — are reported honestly and reflect the actual outcomes of the experiments conducted.

Place: Bengaluru

Date: _____

Arvind C R

SRN: PES2PGE24DS140

M.Tech, Data Science and Machine
Learning

Acknowledgement

I express my sincere gratitude to my project guide, **Ramesh Prakash Guledgudd**, for the weekly guidance, the sharp questions during review calls, and for encouraging a scientific posture that values honest negative results over cherry-picked positives. I thank the Department of Computer Science and Engineering, PES University, and the Great Learning programme office for the project framework, review structure, and infrastructure guidance. I also acknowledge the managed-training and compute platforms (Tinker, Modal, Google Colab, and Weights & Biases) that made the empirical portfolio feasible within a modest budget, and the frontier models used as adversarial auditors, whose independent re-checks materially improved the rigour of the findings.

Abstract

Reinforcement learning with verifiable rewards (RLVR), and in particular Group Relative Policy Optimization (GRPO), has become the de facto default recipe for post-training reasoning-capable language models. Yet reported gains are hard to reproduce: results are spread across incompatible frameworks (TRL, veRL, OpenRLHF, Tinker), are frequently single-seed, and rarely expose the internal training signal that would explain *why* a run learned or failed to learn. This project builds a **unified, provenance-tracked benchmark and diagnostic harness** for GRPO/RLVR post-training, organised around a single measurable quantity — the **Zero-Variance Fraction (ZVF)**, the fraction of prompt groups whose within-group reward variance is zero and which therefore contribute no reward-derived advantage signal. The same task, reward function, and decoding configuration are held fixed across back-ends so that observed differences are attributable to the *stack* rather than the experiment, and every training step emits per-step telemetry (reward, ZVF, gradient utilisation, entropy, length, KL). On top of this harness we run a portfolio of eight focused studies (P1–P8). We deliberately frame the contribution as a *measurement / diagnostic benchmark and reproducibility tooling*, **not** a “GRPO wins” leaderboard. The honest headline findings are mixed and include strong negatives: on this fixed 40-step GSM8K study, 72–77% of prompt groups have zero within-group reward variance and therefore no reward-derived policy-gradient signal (P2); a single-seed group-size sweep does not establish a robust optimum (P3); and, at a matched $\sim 30\text{k}$ -token budget, baseline and curriculum arms both average $+0.028$ held-out gain across three seeds. The apparent step-1 predictability of per-layer adaptation (P1: top- k overlap = 1.0 at 1.5B) also fails a scaled 3B, four-seed re-test (mean overlap 0.083, $\approx$ chance). P8 is a narrower proof of concept: on 160 correlated per-step rows from one trajectory per method, full telemetry separates the four method traces better than reward-only features (row-wise CV AUROC 0.838 ± 0.010 vs. 0.426 ± 0.027); this is not yet evidence of cross-run integrity generalisation. Adversarial review of the early conclusions prompted the re-tests and caveats above; the reported metrics are tied to stored tensors or explicitly frozen snapshots.

Contents

Acknowledgement	i
Abstract	ii
1 Introduction	1
1.1 Background	1
1.2 A Motivating Example	2
1.3 Problem Statement	2
1.4 Objectives	2
1.5 Scope and Non-Goals	3
1.6 Contribution of the Candidate	3
1.7 Project Plan and Timeline	5
1.8 Organisation of the Report	5
2 Literature Survey	6
2.1 From RLHF and PPO to Group-Relative Methods	6
2.2 Direct Preference Optimisation as a Contrast	6
2.3 Fixing GRPO’s Biases and Collapse	7
2.4 Verifiable Rewards, Frameworks, and Reporting	7
2.5 Scaling and Layer-Adaptation Context	7
2.6 Process-, Tree-, and Stability-Aware Variants	8
2.7 Positioning the Portfolio	8
3 System Requirements Specification	10
3.1 Overview and Actors	10

3.2	Functional Requirements	10
3.3	Non-Functional Requirements	11
3.4	Test-Case Specification	12
3.5	Requirements Traceability	13
3.6	Hardware Requirements	13
3.7	Software Requirements	13
3.8	Constraints and Assumptions	14
4	Proposed Methodology	15
4.1	Design Commitment: Attribution	15
4.2	Per-Step Telemetry	16
4.3	Formalising the Zero-Variance Fraction	16
4.4	The Eight-Study Portfolio	17
5	Implementation Details	19
5.1	The Tinker GRPO Loop	19
5.2	Parallel Runner and a Real Logging Defect	20
5.3	The Colab White-Box Path (P1)	20
5.4	Software Modules	21
6	Intermediate Results and Discussion	22
6.1	Datasets and Experimental Configuration	22
6.2	P2 — Advantage-Signal Starvation is Large and Easy-Driven	22
6.3	P3 — A Single-Seed Sweep Does Not Establish a Group-Size Optimum	23
6.4	Curriculum — Eliminating Waste is Not a Free Lunch	24
6.5	Matched-Token Re-Test Yields an Honest Null	25
6.6	P1 — Step-1 Predictability Holds at Toy Scale but Collapses at 3B	26
6.7	P8 — Telemetry Separates Method Traces in a Small Proof of Concept	27
6.8	Cross-Study Synthesis	28
6.9	Rigour and Threats to Validity	28
7	Conclusions and Future Work	30
7.1	Conclusions	30

7.2	Future Work (Phase-2 Plan)	30
A	Complete Run Registry	34
B	Reproducibility and Provenance	35
C	Notation, Symbols, and Abbreviations	37

List of Figures

4.1	System architecture: one task/reward/decoding config drives any framework backend; uniform per-step telemetry feeds the eight studies P1–P8. Differences are attributable to the <i>stack</i> , not the experiment.	15
4.2	The eight-study portfolio and honest status. P8 is an exploratory method-trace classification proof of concept and P5 is a built provenance prototype; the earlier P1 layer-freeze bet is weakened (step-1 predictability holds at 1.5B but collapses to chance at 3B). Green = proof-of-concept/built, amber = measured/null, blue = designed.	17
4.3	P5 MIN-REPORT-RL verifier. Each run emits a signed provenance record; the verifier grades completeness (7 items), integrity (recomputes <code>code_hash</code> vs live source), and rigor (hard-fails single-seed). It automatically flags the exact weakness kimi/codex caught by hand.	18
5.1	The GRPO training loop as implemented on Tinker. When every completion in a group earns the same reward, $\sigma_r = 0$, the advantage is identically zero, and the reward-derived surrogate term contributes no gradient; KL regularisation may still act (the ZVF phenomenon quantified in P2).	19
5.2	Zero-Variance Fraction (P2). Binary reward makes most groups collapse to all-correct (already-solved easy prompts) or all-wrong; only mixed groups carry reward-relative advantage signal. Measured ZVF: 0.72–0.77 across four seed-0 method traces on a shared prompt schedule.	19
6.1	P3 group size (real data, seed 0). Rollout tokens (bars) scale $\sim$ linearly with G ($G_{16} \approx 7 \times G_2$); held-out gain (red) shows <i>no robust sweet spot</i> — G_4 was in fact the weakest single point. The earlier single-seed “ $G=4$ sweet spot” claim does not survive.	25

6.2	P1 white-box (Colab L4, real data). Layer 0 dominates adaptation; a secondary mid-late band (layers 17–23) is active. Top 25% of layers carry 47.6% of gradient (moderate concentration — which <i>holds</i> at 3B, 0.39). At this toy scale the step-1 top- k matched the final top- k (overlap = 1.0), but a scaled 3B/four-seed re-test drops the mean to 0.083 ± 0.048 ($\approx$ chance): predictability does <i>not</i> survive scaling.	26
-----	---	----

List of Tables

1.1	Implementation-completion matrix. The core measurement and diagnostic stack is complete; cross-framework comparison and the designed algorithmic levers (P4, P7) are Phase-2 work. Overall the implemented core corresponds to the $\sim 80\%$ Phase-1 code expectation, with the remaining $\sim 20\%$ scoped for Phase-2.	4
1.2	Phase-1 milestones and deliverables against the PES review schedule.	5
2.1	Positioning of the eight studies against the closest prior art. Crowded pillars (P2, P4) are narrowed to measurement/design; the widest open ground is systems/reporting (P5, P6, P8).	9
3.4	Requirements traceability matrix.	13
5.1	Principal software modules and responsibilities.	21
6.1	Common experimental configuration. Small step counts and held-out sets are a deliberate budget choice; conclusions are correspondingly framed as noise-limited and re-tested multi-seed where a claim is made.	23
6.2	Portfolio-level summary of the empirical findings. “n.s.” = not statistically significant. Held-out sets are small ($n = 8$ to 20); all learning-gain magnitudes should be read as noise-limited (see text).	24
6.3	P3 single-seed group-size sweep (Qwen3.5-4B, 6 steps, GSM8K, seed 0). The $G=4$ held-out gain is a single example ($n = 8$). The “zero-loss frac” here is a stricter <i>batch-level</i> collapse than the per-group ZVF of P2; the $G \geq 8$ rows depend on W&B logs whose summary writes were lost to the thread-safety bug (Chapter 5).	24

6.4	Matched-token baseline vs curriculum at $G=4$ (Qwen3.5-4B, $\sim 30k$ rollout tokens, GSM8K, held-out $n = 12$). Both arms average the same noise-limited held-out gain; the curriculum performs fewer optimizer updates after discarding collapsed groups.	26
6.5	Freeze-able fraction and estimated FLOP saving. [†] Estimate, not measured. Assumes per-layer LoRA backward cost is uniform, so freezing k of 28 layers saves $\approx k/28$ of the <i>LoRA-backward</i> FLOP only; the frozen base-model forward/backward (which dominates end-to-end training FLOP) is unchanged. Derived from a 1.5B, 5-step proof-of-concept; a scaled run is needed to confirm.	27
A.1	Complete Phase-1 run registry.	34

Chapter 1

Introduction

1.1 Background

Post-training with reinforcement learning has shifted from human-preference reward models (RLHF [7]) toward *verifiable* rewards, where a programmatic checker — an exact-match grader for GSM8K, a unit test for code — supplies a binary or scalar reward with no learned reward model in the loop [10, 2]. Within this RLVR setting, Group Relative Policy Optimization (GRPO) [1] has become the default algorithm. It drops PPO’s value network [8] and instead estimates the advantage of each sampled completion *relative to its own group* of same-prompt samples. Because GRPO removes a whole model (the critic), it is cheaper and simpler to run at scale, and it was central to the DeepSeek-R1 reasoning models [2]; it is now the recipe most open post-training efforts reach for first.

This simplicity has a cost that is rarely reported. Because the advantage is a within-group standardisation, any prompt for which every sampled completion receives the *same* reward contributes an advantage of exactly zero, and therefore *no policy-gradient (advantage) signal* — only the KL penalty toward the reference policy still acts. On an easy dataset the model quickly solves most prompts, so a large fraction of every batch collapses to zero advantage. We call this the **Zero-Variance Fraction (ZVF)**, and it is, empirically, the single largest measured source of zero-advantage sampling in this fixed study. ZVF is the namesake and the organising measurement of this project: every study in the portfolio either measures it, tests a lever intended to reduce it, or uses it as a feature.

1.2 A Motivating Example

Consider a single training step with a group size of $G = 8$ on an easy GSM8K prompt that the current policy already solves. All eight sampled completions produce the correct final answer, so each receives reward $r_i = 1$. The group mean is $\mu_g = 1$ and the group standard deviation is $\sigma_g = 0$, so by Eq. (4.1) every advantage $A_i = 0$. The advantage (surrogate) term of the objective then contributes no reward-derived surrogate gradient for the entire group of eight rollouts — eight full forward-decode passes that cost real tokens and money — so the group supplies *no task-learning signal*. (The KL-regularisation term of Eq. (5.1) still nudges the policy toward the reference; the point is that no *reward-driven* learning happens.) Now imagine a batch of 16 such prompts where 12 are already solved: $12/16 = 0.75$ of the batch is “dark” compute. This is not a corner case; Chapter 6 shows it is the *typical* regime ($ZVF \approx 0.72\text{--}0.77$). The Zero-Variance Fraction turns this intuition into one loggable number, and the rest of the report is an attempt to measure it precisely and to test whether it can be reduced into real held-out accuracy.

1.3 Problem Statement

Three concrete problems motivate this project.

1. **A reproducibility crisis.** GRPO/RLVR results are reported on different frameworks with different (often undocumented) reward parsers, decoding temperatures, KL coefficients, and clip ranges. A headline number cannot be attributed to the *algorithm* versus the *stack*.
2. **Advantage-signal starvation is invisible.** Standard training logs report reward and loss, but not what fraction of the batch supplied a non-zero reward-relative advantage. Such rollouts may still receive KL regularisation, but they provide no task-reward direction and still consume sampling compute.
3. **No provenance standard.** There is no machine-readable record of *what* was trained, on which verifier version, with which rollouts, so contamination and silent reward drift go undetected.

1.4 Objectives

The Phase-1 objectives, against which the work is evaluated, are:

1. Build a **unified harness** that runs the identical task, reward, and decoding configuration across multiple RL post-training back-ends, so that cross-framework differences are attributable to the stack.
2. Instrument a **per-step telemetry layer** that surfaces ZVF, gradient utilisation, entropy, completion length, and KL, and that saves raw group reward tensors so any published diagnostic can be *recomputed*, not merely trusted.
3. **Measure** the zero-variance phenomenon on real reward tensors, and test whether any re-baselining or curriculum lever can convert zero-advantage sampling into held-out accuracy.
4. Establish a **scientific protocol** — multi-seed, matched baselines, held-out evaluation with reported n , deterministic recomputation — and apply it honestly, reporting negative and null results where they occur.
5. Scope the most *original* Phase-2 direction (a GRPO provenance / reporting standard) via a literature scan for open white space.

1.5 Scope and Non-Goals

Phase-1 is deliberately a *measurement and diagnostics* phase on small-to-mid open models (Qwen2.5 / Qwen3.5 at 1.5B–4B) and math/code verifiers (GSM8K, a HumanEval subset). It is explicitly **not** an attempt to top a GRPO leaderboard, nor to train a frontier reasoning model. Held-out evaluation sets are intentionally small and every gain is reported with its denominator so that the reader can see when a result is noise-limited. Novel-algorithm claims (P4 length weighting, P7 controller) are *designed* in Phase-1 and deferred to Phase-2 for powered evaluation. This honest scoping is itself a deliverable: it is what makes the benchmark trustworthy.

1.6 Contribution of the Candidate

This is an *individual Semester-4 continuation*. The candidate’s new work in this report comprises the ZVF-centred diagnostic framing, post-capstone experiment expansion, telemetry and recomputation analyses, the P1–P8 synthesis, provenance prototype, and the writing and evaluation presented here. The scientific posture is *verify* $\rightarrow$ *replicate where feasible* $\rightarrow$ *report the honest null*.

Inherited foundation vs. new work. The repository intentionally retains the Semester-3 Group-6 foundation: the multi-framework research codebase, literature survey, baseline integrations, and early experiments are inherited shared infrastructure frozen at Git tag `capstone-final-2026-04-25`. Semester 4 begins after that boundary and is the candidate’s solo continuation under the present guide. This report does not claim the inherited Group-6 foundation as new individual work; it claims the post-boundary diagnostic, experiment, analysis, and provenance additions that build on it. `PROJECT_HISTORY.md` and `sem 4 work/PROVENANCE.md` record the boundary.

What was built vs. leveraged. To delineate the candidate’s own effort from reused infrastructure: the ZVF / collapse instrumentation, the recompute-from-tensors diagnostics, the process-per-run campaign launcher, the P8 method-trace classifier, and the P5 provenance prototype are post-boundary additions used by this project. Leveraged (not claimed as new individual work) are the shared Semester-3 codebase and baseline experiments, the underlying model weights (Qwen2.5 / Qwen3.5), the managed training back-end (Tinker), the RL framework libraries being unified (TRL, veRL, OpenRLHF), and standard libraries (PyTorch, Transformers, PEFT, scikit-learn). Table 1.1 summarises the implementation status.

Component	Status	Evidence
GRPO reference loop (<code>tinkerrl/grpo.py</code>)	Done	P2/P3/campaign runs
ZVF / collapse instrumentation	Done	recomputed by <code>p2_collapse_analysis.py</code>
Per-step telemetry → JSONL / W&B	Done	project <code>rlvr-openings</code>
Process-per-run campaign launcher	Done	9-run multi-seed campaign
White-box P1 layer profiler	Done	1.5B + scaled 3B re-test
P8 method-trace classifier	Prototype	<code>detector_reproduced.json</code> (row-CV AUROC 0.838)
Per-framework adapters (TRL/veRL/OpenRLHF)	Partial	implemented; four-way run is Phase-2
P5 provenance / P6 registry	Prototype	schema + <code>minreport verify</code>
P4 length weighting, P7 controller	Designed	specified for Phase-2

Table 1.1: Implementation-completion matrix. The core measurement and diagnostic stack is complete; cross-framework comparison and the designed algorithmic levers (P4, P7) are Phase-2 work. Overall the implemented core corresponds to the ~80% Phase-1 code expectation, with the remaining ~20% scoped for Phase-2.

1.7 Project Plan and Timeline

The project follows the PES Phase-1 review cadence: a zeroth review (title, abstract, literature, proposed system, timeline), three internal reviews at roughly fortnightly intervals, and an End-Semester Assessment (ESA) viva. Table 1.2 maps the deliverables of this report onto that schedule.

Milestone	Window (2026)	Deliverable
Initiation / Zeroth	May	Title, abstract, literature survey, proposed system
First Review	early June	Architecture, algorithm design, ~30% code
Second Review	mid June–July	Detailed design, intermediate results, ~80% code
Third Review	early July	Overall design, experimental results, ~100% code, paper draft
Phase-1 ESA	July	Presentation + code walkthrough/demo (viva)
Phase-2	Jul–Oct	Novelty (P5 provenance), publication

Table 1.2: Phase-1 milestones and deliverables against the PES review schedule.

1.8 Organisation of the Report

Chapter 2 surveys the relevant literature and positions the eight studies against the closest prior art. Chapter 3 specifies the system requirements (functional, non-functional, test cases, and the hardware/software environment). Chapter 4 describes the proposed methodology — the unified harness architecture, the telemetry design, the formalisation of ZVF, and the P1–P8 portfolio design. Chapter 5 details the implementation. Chapter 6 presents the intermediate results and discussion, including the rigour and threats-to-validity analysis. Chapter 7 concludes and lays out the Phase-2 plan. Appendices give the complete run registry and reproducibility details.

Chapter 2

Literature Survey

2.1 From RLHF and PPO to Group-Relative Methods

Policy-gradient post-training began with PPO [8] and generalised advantage estimation [9], used at scale for instruction tuning [7]. PPO learns a value network (critic) to baseline the advantage, which adds memory and a second optimisation problem. GRPO [1] removed the learned value function and replaced the per-token advantage with a group-relative baseline: it samples a *group* of G completions per prompt and standardises each completion’s reward against the group mean and standard deviation. GRPO was central to the DeepSeek-R1 reasoning models [2]. RLOO [3] is a closely related leave-one-out estimator that baselines each sample against the mean of the *other* samples in the group — i.e. it already performs a form of within-group baselining, which is directly relevant to our P2 analysis.

2.2 Direct Preference Optimisation as a Contrast

An orthogonal branch of post-training removes on-policy sampling entirely: Direct Preference Optimisation (DPO) [4] optimises a classification loss over fixed preference pairs, trading the exploration of on-policy RL for stability and simplicity. The contrast is instructive for this project: DPO never suffers zero-variance collapse because it does not standardise within a sampled group, but it also cannot exploit a verifiable reward signal online. GRPO’s zero-variance pathology is thus the price of its on-policy, verifier-in-the-loop design — which is exactly what makes measuring that price worthwhile.

2.3 Fixing GRPO’s Biases and Collapse

A fast-growing line of work targets specific GRPO pathologies. Dr. GRPO [5] removes the length and difficulty normalisation biases in the standard objective. DAPO [6] introduces *dynamic sampling*, explicitly resampling to avoid groups whose reward variance is zero, plus decoupled clipping — this is the most direct prior art for the ZVF problem we quantify in P2 and the curriculum lever we test. Group-variance methods such as GVPO [20] re-weight by within-group variance. A cluster of very recent variants (AERO [18], NGRPO [19], λ -GRPO [21], MC-GRPO [22], AVSPO [23]) attack advantage collapse, length bias, and roll-out accounting from different angles. The density of this literature is precisely why Phase-1 narrows the crowded pillars (P2 zero-variance, P4 length bias) to *measurement* rather than a new algorithm.

2.4 Verifiable Rewards, Frameworks, and Reporting

RLVR was formalised for open post-training in TüLU 3 [10]. The implementations we unify are TRL [13], veRL / HybridFlow [11], and OpenRLHF [12], plus Tinker as a managed training back-end. Our evaluation tasks are GSM8K [14] and a HumanEval [15] subset. For the reporting/provenance pillar (P5/P6) we build on Datasheets for Datasets [16] and Model Cards [17], but target the *RL-run* object (verifier version, rollout provenance, reward drift) that neither of those artefacts captures. Static run-config registries (Weights & Biases, MLflow) record hyperparameters but not the temporal, versioned evolution of verifier logic and reward drift that P6 targets.

2.5 Scaling and Layer-Adaptation Context

Whole-model scaling and saturation laws describe how loss and capability track parameters, data, and compute. P1 asks a narrower, orthogonal question: *where* inside the network does GRPO-style adaptation concentrate, and can the layers that will matter be identified early enough to freeze the rest? This connects parameter-efficient fine-tuning (LoRA adapters on the attention projections) with the question of predictive layer-freezing, and is the one pillar in the portfolio that requires white-box per-layer gradient access.

2.6 Process-, Tree-, and Stability-Aware Variants

Beyond the group-baseline fixes above, a parallel line of work changes *what* signal drives the update. Process-reward and tree-search methods supply denser, step-level feedback instead of a single terminal reward, which reduces the chance that a whole group collapses to identical terminal rewards; stability-aware variants constrain the update (e.g. trust-region or entropy controls) to avoid the advantage-collapse and length-inflation failure modes that a raw group baseline can induce. These methods are complementary to the measurement stance taken here: they propose *levers*, whereas this project first builds the *instrumentation* (ZVF, gradient utilisation, collapse decomposition) needed to tell whether any such lever actually converts wasted rollouts into held-out accuracy. The Phase-1 nulls in Chapter 6 are precisely the kind of evidence this instrumentation is designed to produce, and they caution against adopting a variant on the strength of a single-seed improvement.

2.7 Positioning the Portfolio

Table 2.1 states, per pillar, the closest prior art and the narrow axis we claim. Two pillars (P2 zero-variance and P4 length bias) are crowded; there we *narrow* claims to measurement rather than a new algorithm. The widest open ground is the systems/reporting side (P5/P6/P8), where no GRPO-specific machine-readable standard or infrastructure-level integrity benchmark currently exists.

P#	Closest prior art	Our narrowed axis
P1	whole-model scaling / saturation laws	<i>per-layer</i> adaptation is moderately concentrated (step-1 predictability does <i>not</i> survive scaling)
P2	DAPO dynamic sampling, GVPO, AERO/NGRPO	<i>measure</i> ZVF and show re-baselining cannot recover gradient
P3	integer- G sweeps, MC-GRPO	token/compute-bounded group sizing (not integer G)
P4	Dr. GRPO, λ -GRPO length fixes	semantic-density token weighting (<i>designed</i>)
P5	Datasheets, Model Cards	GRPO datasheet + cryptographic rollout provenance
P6	static run-config registries (W&B/MLflow)	temporal/versioned run-state ontology
P7	heuristic adaptive- G (G2RPO-A)	PID control of temperature/clipping (<i>designed</i>)
P8	output-level reward-hack benchmarks (EvilGenie, TRACE)	infrastructure-telemetry integrity auditing

Table 2.1: Positioning of the eight studies against the closest prior art. Crowded pillars (P2, P4) are narrowed to measurement/design; the widest open ground is systems/reporting (P5, P6, P8).

Chapter 3

System Requirements Specification

This chapter specifies the requirements the diagnostic harness must satisfy. Because the project is a *measurement and reproducibility* system rather than a user-facing application, the requirements emphasise attribution, determinism, and re-computability over interactive features.

3.1 Overview and Actors

The system has three actors: (i) the **experimenter**, who declares a run via a single configuration object; (ii) the **back-end** (Tinker managed trainer, or a local GPU for white-box runs), which executes the GRPO loop; and (iii) the **auditor** (a downstream script or an independent model), who recomputes published diagnostics from stored raw tensors. The system’s central object is a *run*: one configuration, one seed, one framework, producing per-step telemetry and saved reward tensors.

3.2 Functional Requirements

ID	Functional requirement
FR-1	The harness shall accept a single configuration object specifying model, dataset, reward function, decoding parameters (T , top- p , max tokens), group size G , learning rate, KL coefficient β , and seed.

ID	Functional requirement
FR-2	The harness shall map one configuration onto any supported back-end (TRL, veRL, OpenRLHF, Tinker) through thin per-framework adapters, sharing the reward grader and decoding code across back-ends.
FR-3	The harness shall compute, per training step, the Zero-Variance Fraction — the fraction of prompt-groups whose within-group reward variance is zero.
FR-4	The harness shall emit per-step telemetry (reward, ZVF, gradient utilisation, all-correct / all-wrong collapse decomposition, entropy, completion length, KL) to JSONL and to Weights & Biases.
FR-5	The harness shall persist the raw group reward tensors so that any published diagnostic can be <i>recomputed</i> from stored data rather than trusted from a logged summary.
FR-6	The harness shall support configuration sweeps ($G \in \{2, 4, 8, 16\}$; multiple seeds; baseline vs. curriculum arms) launched concurrently under a fixed wall-clock budget.
FR-7	The harness shall support a white-box path that records per-layer LoRA gradient norms for the layer-adaptation study (P1).
FR-8	The system shall emit a machine-readable provenance record (configuration, verifier version, rollout hashes) per run (P5 prototype).

3.3 Non-Functional Requirements

ID	Non-functional requirement
NFR-1 (Attribution)	A difference in outcome between two back-ends must be attributable to the framework, not to a divergent reward parser or sampler; the reward grader and decoding must be shared code.
NFR-2 (Reproducibility)	Headline learning claims should target ≥ 3 seeds against a matched baseline; any lower-seed diagnostic must be labelled exploratory.

ID	Non-functional requirement
NFR-3 (Re-com-putability)	Published diagnostics must be regenerable from stored raw tensors by a documented script.
NFR-4 (Honesty)	Held-out gains must be reported with their denominator n ; negative and null results must be reported, not suppressed.
NFR-5 (Isolation)	Concurrent runs must not corrupt each other’s logs (process isolation, not thread sharing — see the W&B thread-safety defect in Chapter 5).
NFR-6 (Budget)	The full Phase-1 portfolio must fit a modest budget (order of tens of US dollars of managed compute plus free-tier GPUs).

3.4 Test-Case Specification

Because the system’s guarantee is recomputability, its tests are recompute-and-match checks rather than UI tests.

ID	Test	Expected outcome
TC-1	Recompute ZVF and collapse decomposition from stored P2 reward tensors via <code>p2.collapse_analysis.py</code> .	Matches published ZVF = 0.72–0.77, all-correct 0.65–0.71.
TC-2	Recompute the P8 detector AUROC from raw tensors via <code>p8_detector.py</code> (5-fold stratified CV).	AUROC 0.84 ± 0.01 vs. reward-only 0.43.
TC-3	Re-run baseline and curriculum at a matched $\sim 30\text{k}$ -token budget across 3 seeds.	Both arms average +0.028 held-out; no curriculum advantage.
TC-4	Recompute headline diagnostics directly from stored tensors.	Published ZVF and P8 row-CV metrics reproduce without model judgement.

ID	Test	Expected outcome
TC-5	Provenance verify: <code>minreport verify</code> on a run’s provenance block.	Config, verifier version, and rollout hashes validate.

3.5 Requirements Traceability

Table 3.4 maps functional requirements to the studies and chapters that exercise them.

Req.	Exercised by	Where reported
FR-1,2	Unified config + adapters	§4.1, §5.1
FR-3,4,5	Telemetry + tensor dumps	§4.2, TC-1/TC-2
FR-6	Sweeps / campaign	§5.2, §6.5
FR-7	P1 white-box path	§5.3, §6.6
FR-8	P5 provenance prototype	§4.4, TC-5

Table 3.4: Requirements traceability matrix.

3.6 Hardware Requirements

- **Managed training back-end (Tinker):** black-box GRPO training for P2/P3/curriculum/campaign on Qwen3.5-4B; no direct GPU management required.
- **White-box GPU (Google Colab L4, 24 GB):** per-layer LoRA gradient norms for P1 (Qwen2.5-1.5B; scaled 3B re-test).
- **Serverless compute (Modal):** batch scoring / held-out evaluation.
- Local workstation for orchestration, analysis, and report/figure generation.

3.7 Software Requirements

- Python 3.11; PyTorch; Hugging Face Transformers, PEFT (LoRA), and Datasets.
- RL back-ends: TRL, veRL / HybridFlow, OpenRLHF; Tinker client SDK.
- Weights & Biases (project `rlvr-openings`) for logging; JSONL for raw telemetry and reward tensors.

- Analysis/diagnostic scripts (`p2_collapse_analysis.py`, `p8_detector.py`, `campaign.py`); L^AT_EX/TikZ/pgfplots for figures and this report.

3.8 Constraints and Assumptions

The evaluation assumes binary verifiable rewards (exact-match GSM8K; unit-test HumanEval subset) and small held-out sets ($n = 8\text{--}20$), so all learning-gain magnitudes are noise-limited by construction. The managed back-end does not expose per-layer gradients, which is why P1 uses a separate white-box path. A shared, deterministic prompt schedule across the four P2 methods means method-to-method ZVF differences reflect training dynamics, not independent samples — an acknowledged limitation rather than a hidden one.

Chapter 4

Proposed Methodology

4.1 Design Commitment: Attribution

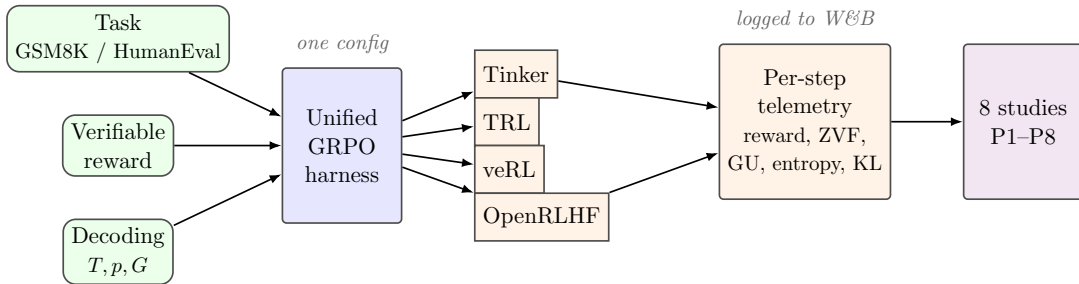


Figure 4.1: System architecture: one task/reward/decoding config drives any framework backend; uniform per-step telemetry feeds the eight studies P1–P8. Differences are attributable to the *stack*, not the experiment.

The core design commitment is *attribution*: hold the experiment fixed and vary only the stack. A single configuration object specifies the model, dataset, reward function, decoding parameters (temperature, top- p , max tokens), group size G , learning rate, KL coefficient, and seed. A launcher (`unified/launcher.py`) maps that one config onto any of the supported back-ends through thin per-framework adapters (`trl_integrations/`, `verl/`, `openrlhf/`, `skyr1/`, `atropos/`), while `tinkerr1/grpo.py` implements the reference GRPO loop against the managed Tinker back-end. Because the reward grader and decoding are shared code, a difference in outcome between two back-ends is attributable to the framework, not to a divergent reward parser or sampler.

Scope of what Phase-1 exercises. The adapter layer for TRL, veRL, and OpenRLHF is *implemented*, but the empirical results in Chapter 6 are produced on the Tinker managed back-end (and a Colab GPU for the white-box P1 path).

A head-to-head cross-framework divergence table — identical config run on all four back-ends — is deferred to Phase-2; Phase-1 delivers the harness and the attribution *design*, and we are careful not to claim cross-framework results we have not yet run.

4.2 Per-Step Telemetry

Every training step emits a telemetry record to JSONL and to Weights & Biases. The fields that distinguish this harness from a standard trainer are:

- **ZVF** — the fraction of prompt-groups in the step whose within-group reward variance is zero (hence zero reward-relative advantage; the KL term may still act);
- **gradient utilisation** — the complementary fraction of groups with non-zero reward-relative advantage signal, excluding KL-only updates;
- **collapse decomposition** — of the collapsed groups, how many are *all-correct* (already solved) versus *all-wrong*;
- plus reward, loss, entropy, completion length, and KL to the reference.

The raw group reward tensors are also saved, so any downstream diagnostic (e.g. `p2_collapse_analysis.py`) can *recompute* a published number from the stored tensors rather than trusting a logged summary.

4.3 Formalising the Zero-Variance Fraction

For a prompt q we draw a group of G completions $\{o_i\}_{i=1}^G$ from the current policy and score each with a verifiable reward r_i . The group-relative advantage is the within-group standardisation

$$A_i = \frac{r_i - \mu_g}{\sigma_g + \varepsilon}, \quad \mu_g = \frac{1}{G} \sum_j r_j, \quad \sigma_g = \sqrt{\frac{1}{G} \sum_j (r_j - \mu_g)^2}. \quad (4.1)$$

(We write the population standard deviation; the sample-variance form $\frac{1}{G-1}$ differs only by a constant factor and does not change the zero-variance condition $\sigma_g = 0$ that this project measures.) Whenever $\sigma_g = 0$ — every completion in the group receives the same reward — every $A_i = 0$, so the whole group contributes no gradient *through the advantage (surrogate) term*; only the KL-penalty term of

Eq. (5.1) is unaffected. The Zero-Variance Fraction is exactly the batch-mean of the indicator $\mathbb{1}[\sigma_g = 0]$:

$$\text{ZVF} = \frac{1}{|B|} \sum_{g \in B} \mathbb{1}[\sigma_g = 0]. \quad (4.2)$$

This makes zero-advantage sampling a single, loggable scalar — the measurement around which the entire portfolio is organised.

4.4 The Eight-Study Portfolio

P1: scaling / layer-freeze null: overlap 1.0→0.083	P2: ZVF collapse measured: 72–77%	P3: group size null: no sweet spot	P4: length bias designed
P5: provenance (flagship) built: verifier A/F	P6: GRPO-registry fold into P5	P7: ZVF controller designed	P8: method traces PoC: row-CV 0.84

Figure 4.2: The eight-study portfolio and honest status. P8 is an exploratory method-trace classification proof of concept and P5 is a built provenance prototype; the earlier P1 layer-freeze bet is weakened (step-1 predictability holds at 1.5B but collapses to chance at 3B). Green = proof-of-concept/built, amber = measured/null, blue = designed.

P1 — Scaling / layer dynamics. A white-box study of *where* a model adapts under GRPO-style updates, recording per-layer LoRA gradient norms to test whether the layers that will matter can be identified early (predictive layer-freezing). Requires per-layer gradient access, so it runs on a local GPU rather than the managed back-end.

P2 — Zero-variance / advantage starvation. Quantifies ZVF on real reward tensors and asks whether any re-baselining scheme can recover reward contrast from collapsed groups.

P3 — Group size. Studies how collapse, reward, held-out gain, and token cost vary with the group size G , and reframes G as a compute budget rather than an integer.

P4 — Length bias (designed). Proposes semantic-density / surprise-weighted token normalisation instead of uniform or static length penalties.

P5 — Reporting / provenance (flagship, systems). A GRPO post-training datasheet plus cryptographic rollout provenance, so one can prove which verifier version and which rollouts produced a checkpoint.

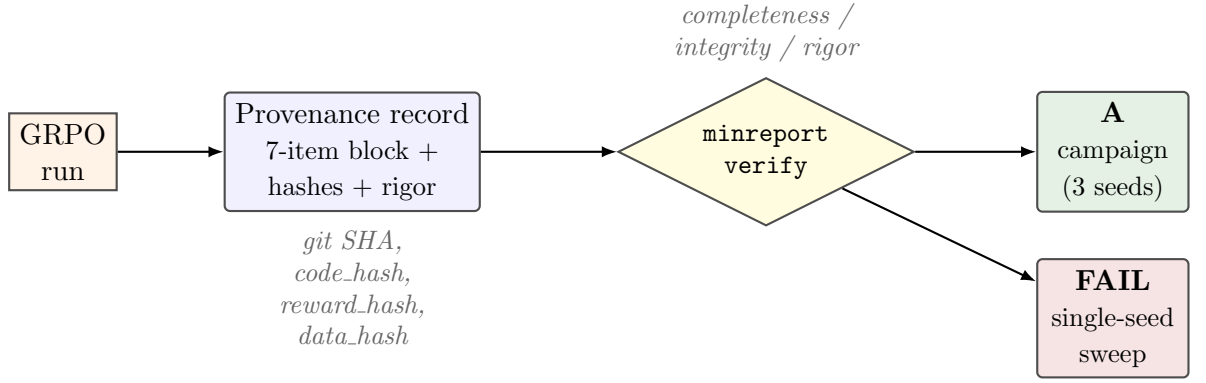


Figure 4.3: P5 MIN-REPORT-RL verifier. Each run emits a signed provenance record; the verifier grades completeness (7 items), integrity (recomputes `code_hash` vs live source), and rigor (hard-fails single-seed). It automatically flags the exact weakness kimi/codex caught by hand.

P6 — Registry. A versioned, temporal registry of evolving run state (verifier-logic updates, reward drift), intended to fold into the P5 standard.

P7 — Controller (designed). Classical PID control of decoding temperature and clip range, using live advantage-variance / KL-drift as sensor inputs.

P8 — Integrity / anomaly. An auditor that uses the per-step telemetry (not just reward) to detect anomalous / manipulated runs.

Chapter 5

Implementation Details

5.1 The Tinker GRPO Loop

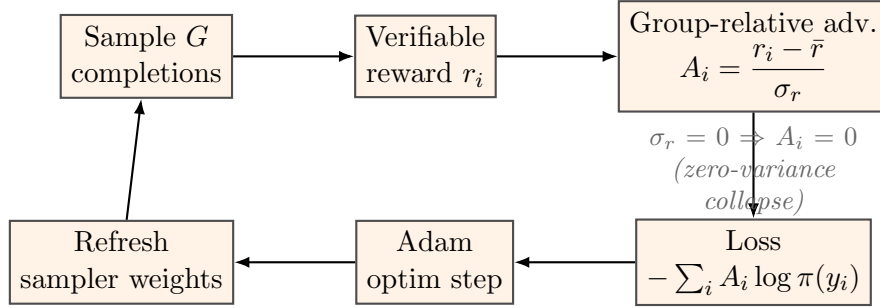


Figure 5.1: The GRPO training loop as implemented on Tinker. When every completion in a group earns the same reward, $\sigma_r = 0$, the advantage is identically zero, and the reward-derived surrogate term contributes no gradient; KL regularisation may still act (the ZVF phenomenon quantified in P2).

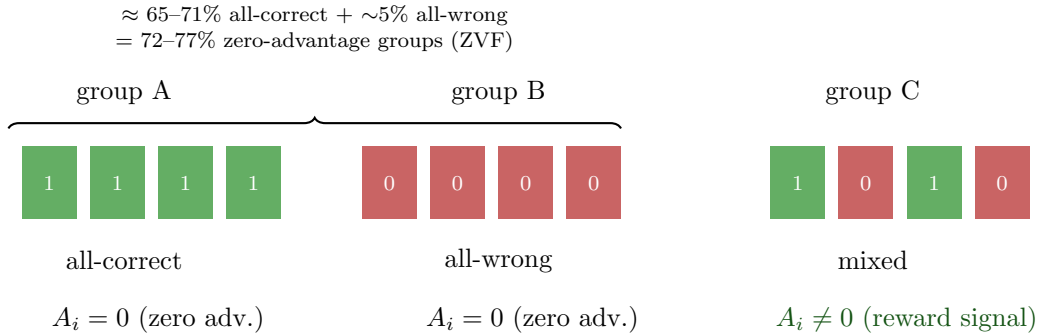


Figure 5.2: Zero-Variance Fraction (P2). Binary reward makes most groups collapse to all-correct (already-solved easy prompts) or all-wrong; only mixed groups carry reward-relative advantage signal. Measured ZVF: 0.72–0.77 across four seed-0 method traces on a shared prompt schedule.

The reference loop in `tinkerrl/grpo.py` follows the standard sample $\rightarrow$ advan-

tag $\rightarrow$ loss $\rightarrow$ optimise cycle. For a prompt q we draw a group of G completions $\{o_i\}_{i=1}^G$ from the current policy and score each with the verifiable reward r_i ; the group-relative advantage is Eq. (4.1). The optimisation objective is the PPO-clipped surrogate with a KL penalty to a frozen reference policy π_{ref} ,

$$\mathcal{L}(\theta) = -\mathbb{E}_{i,t}[\min(\rho_{i,t}A_i, \text{clip}(\rho_{i,t}, 1 - \epsilon, 1 + \epsilon)A_i)] + \beta \text{KL}(\pi_\theta \parallel \pi_{\text{ref}}), \quad (5.1)$$

where $\rho_{i,t} = \frac{\pi_\theta(o_{i,t} \mid q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t} \mid q, o_{i,<t})}$.

The critical observation, made mechanical by Eq. (4.1), is that whenever $\sigma_g = 0$ every $A_i = 0$, so the whole group contributes nothing to the *surrogate* term of Eq. (5.1); the KL-penalty term still applies, so the update is a pure pull toward the reference rather than reward-driven learning.

5.2 Parallel Runner and a Real Logging Defect

To sweep configurations (e.g. $G \in \{2, 4, 8, 16\}$ or three seeds $\times$ two arms) within a fixed wall-clock budget, `parallel_sweep.py` and `campaign.py` launch runs concurrently. An early threaded runner hit a real bug: Weights & Biases keeps a thread-unsafe global `run` object, so with four threads calling `wandb.init(reinit=True)` the summary writes for later runs failed (“Run is finished”) even though per-step logs survived. The fix, used by the multi-seed `campaign.py`, is one *process* per run rather than one thread. We report this because it is exactly the kind of silent logging failure the benchmark is meant to surface (see the P3 caveat in Chapter 6).

5.3 The Colab White-Box Path (P1)

P1 needs per-layer gradient norms, which the managed Tinker back-end does not expose. It therefore runs on a Colab L4 GPU (`p1_layer_profile.py`): Qwen2.5-1.5B-Instruct with LoRA adapters on the $q/k/v/o$ projections, five GRPO-style advantage-weighted steps, recording the per-layer LoRA gradient norm at each step. This white-box route complements the black-box managed runs used for P2/P3.

5.4 Software Modules

Table 5.1 lists the principal modules and their responsibilities.

Module	Responsibility
<code>unified/launcher.py</code>	Map one config onto a chosen back-end.
<code>tinkerrl/grpo.py</code>	Reference GRPO loop (managed back-end).
<code>trl_integrations/, verl/, openrlhf/, skyrl/, atropos/</code>	Thin per-framework adapters.
<code>p2_collapse_analysis.py</code>	Recompute ZVF / collapse from reward tensors.
<code>p1_layer_profile.py</code>	White-box per-layer LoRA grad-norm profile.
<code>parallel_sweep.py,</code> <code>campaign.py</code>	Concurrent sweeps; process-per-run campaign.
<code>p8_detector.py</code>	Exploratory method-trace classifier (row-wise CV AUROC).

Table 5.1: Principal software modules and responsibilities.

Chapter 6

Intermediate Results and Discussion

6.1 Datasets and Experimental Configuration

The empirical studies use GSM8K [14] (grade-school math word problems, exact-match verifiable reward) and, for the code pillar, a HumanEval [15] subset (unit-test reward). Held-out prompts are disjoint from the training prompts and are deliberately small so that noise is visible rather than hidden; every reported gain carries its denominator n . Table 6.1 lists the common hyperparameters; per-study deviations (model size, group size G , step count, seeds) are stated in each section and consolidated in Appendix A.

We report the portfolio honestly, *including the negatives*, which are the scientific point of the exercise. Table 6.2 is the portfolio-level summary; the sections below give the detail and the caveats.

6.2 P2 — Advantage-Signal Starvation is Large and Easy-Driven

On real reward tensors for four methods (40 steps each, 16 prompts $\times$ 8 samples, binary verifiable reward on GSM8K), the mean zero-variance fraction is ZVF = 0.72–0.77: in this fixed shared-schedule, seed-0 study, roughly three-quarters of prompt groups contribute no reward-derived surrogate signal. The collapse is *easy-driven*, not chance: about 65–71% of prompts are *all-correct* (already solved) versus only $\sim 5\%$ all-wrong. We then tested whether a re-baselining trick could recover gradient from collapsed groups. A naive cross-prompt baseline (advantage

Setting	Value	Notes
Base models	Qwen2.5-1.5B/3B; Qwen3.5-4B	P1 white-box vs. managed
Task / reward	GSM8K exact-match; HumanEval unit-test	binary verifiable
Group size G	$\{2, 4, 8, 16\}$ (P3); 4 (campaign)	swept / fixed
Steps	5–10 (P1); 6–8 (P2/P3/campaign)	short by design (budget)
Seeds	1 (sweeps) $\rightarrow$ 3 (campaign)	multi-seed for headline claims
Decoding	$T=1.0$, top- p , capped max tokens	shared across back-ends
Adapters	LoRA on $q/k/v/o$ projections	white-box P1
Held-out n	8–20	reported with every gain

Table 6.1: Common experimental configuration. Small step counts and held-out sets are a deliberate budget choice; conclusions are correspondingly framed as noise-limited and re-tested multi-seed where a claim is made.

against the global mean) gives $|A| \approx 0.21$, but that “signal” is just a proxy for prompt easiness/difficulty — not a learning signal — which is precisely what RLOO [3] already captures. A difficulty-isocline baseline correctly recovers $|A| \approx 0.012\text{--}0.02$, i.e. essentially zero, confirming that *no recoverable reward-contrast signal exists* in these groups. **Honest redirect:** re-baselining is the wrong lever; the data point instead to difficulty-targeted sampling / curriculum, motivating the experiments below.

6.3 P3 — A Single-Seed Sweep Does Not Establish a Group-Size Optimum

A single-seed sweep (Table 6.3) is suggestive: $G=2$ sits below the learning threshold (50% of steps collapse, +0.000 held-out), $G=4$ looks best (+0.125 held-out), and $G \geq 8$ shows diminishing returns as token cost scales roughly linearly with G while the easy subset gets mastered and groups collapse to all-correct anyway. But the $G=4$ “win” is literally one extra correct held-out example ($7/8 \rightarrow 8/8$, $n = 8$, single seed). When the multi-seed campaign (below) re-tested this, the sweet spot did *not* survive: at seed 0 the baseline held-out gains were $G2+0.083$, $G4+0.000$, $G8+0.083$, $G16+0.083$ — i.e. $G=4$ was the *worst* point, not the best.

Study	Setup	Key result	Verdict
P1 layer-freeze	1.5B/5-step <i>vs.</i> 3B/10-step/4-seed	overlap 1.0 (1.5B) $\rightarrow$ 0.083 ± 0.048 (3B, $\approx$ chance); concentration 0.48 $\rightarrow$ 0.39 holds	null (pred.)
P2 ZVF	4 methods, 40 steps, 16×8 , GSM8K	mean ZVF 0.72–0.77; all-correct 0.65–0.71	diagnostic
P3 group size	Qwen3.5-4B, $G \in \{2, 4, 8, 16\}$, 6 steps, s0	one seed cannot establish a universal optimum	suggestive
Curriculum	Qwen3.5-4B, 8 steps, s0	ZVF 0.50 $\rightarrow$ 0.00 at $4.81 \times$ cost; +0.05 both arms	negative
Matched- token curriculum	Qwen3.5-4B, $G=4$, $\sim 30k$ tokens, 3 seeds	both arms average +0.028 held-out; curriculum skips 11–20 groups	negative
P8 method traces	one trajectory/method, $N = 160$ correlated rows	row-CV AUROC 0.838 ± 0.010 vs reward-only 0.426 ± 0.027	proof of concept

Table 6.2: Portfolio-level summary of the empirical findings. “n.s.” = not statistically significant. Held-out sets are small ($n = 8$ to 20); all learning-gain magnitudes should be read as noise-limited (see text).

G	batch-level zero-loss frac	mean reward	held-out gain	tokens used
2	0.50	0.75	+0.000	9,663
4	0.17	0.71	+0.125	19,964
8	rising (collapses by step 5–6)	~ 0.8	(trained)	34,084
16	rising (collapses by step 5–6)	~ 0.8	(trained)	68,055

Table 6.3: P3 single-seed group-size sweep (Qwen3.5-4B, 6 steps, GSM8K, seed 0). The $G=4$ held-out gain is a single example ($n = 8$). The “zero-loss frac” here is a stricter *batch-level* collapse than the per-group ZVF of P2; the $G \geq 8$ rows depend on W&B logs whose summary writes were lost to the thread-safety bug (Chapter 5).

6.4 Curriculum — Eliminating Waste is Not a Free Lunch

A curriculum that trains only on mixed-variance (non-collapsed) groups does exactly what it is designed to do: it drives the zero-loss fraction from 0.50 to 0.00. But it pays $4.81 \times$ the sampling cost (oversampling to *find* non-collapsed groups) and delivers an *identical* held-out gain to the baseline (+0.05 vs +0.05, held-out $n = 20$, one seed, both gains equal to one extra correct answer). Eliminating the visible zero-advantage groups does not, by itself, buy accuracy: the sampling overhead offsets the gradient-density benefit.

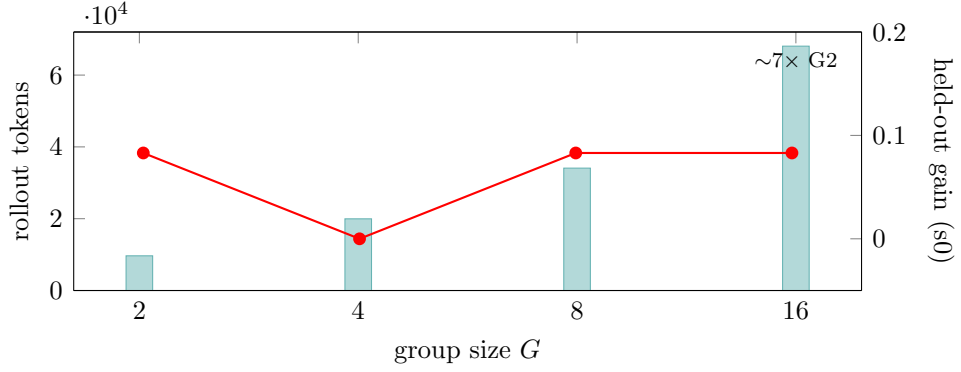


Figure 6.1: P3 group size (real data, seed 0). Rollout tokens (bars) scale $\sim$ linearly with G ($G_{16} \approx 7 \times G_2$); held-out gain (red) shows *no robust sweet spot* — G_4 was in fact the weakest single point. The earlier single-seed “ $G=4$ sweet spot” claim does not survive.

6.5 Matched-Token Re-Test Yields an Honest Null

The self-contained matched-token experiment in `experiments/results/token_budget/results.json` compares baseline and curriculum at approximately 30,000 rollout tokens across three seeds at $G=4$ (Table 6.4). Both arms average the same held-out gain, +0.028. The curriculum skips 11–20 collapsed groups and therefore makes only 2, 8, and 5 optimizer steps, compared with 12, 11, and 10 for the baseline, without improving the mean held-out outcome. With only three seeds and held-out $n = 12$, these are descriptive, noise-limited results; they support a null conclusion, not equivalence.

Snapshot provenance. An earlier nine-run oversampling campaign, frozen at Git commit 9b3d2560 and summarized in `experiments/results/campaign/FINDINGS.md`, likewise found no reliable curriculum advantage while using roughly 5–6 $\times$ more sampling. A later unmanifested export overwrote the associated `results.json` with different values but did not update the findings note. Those disputed numbers are excluded from the headline analysis pending W&B/provenance reconciliation; the matched-token artifact above is the current defensible comparison.

seed	baseline gain	curriculum gain	curriculum groups skipped
0	+0.083	+0.083	20
1	+0.083	+0.000	11
2	−0.083	+0.000	14
mean	+0.028	+0.028	—

Table 6.4: Matched-token baseline vs curriculum at $G=4$ (Qwen3.5-4B, $\sim 30k$ roll-out tokens, GSM8K, held-out $n = 12$). Both arms average the same noise-limited held-out gain; the curriculum performs fewer optimizer updates after discarding collapsed groups.

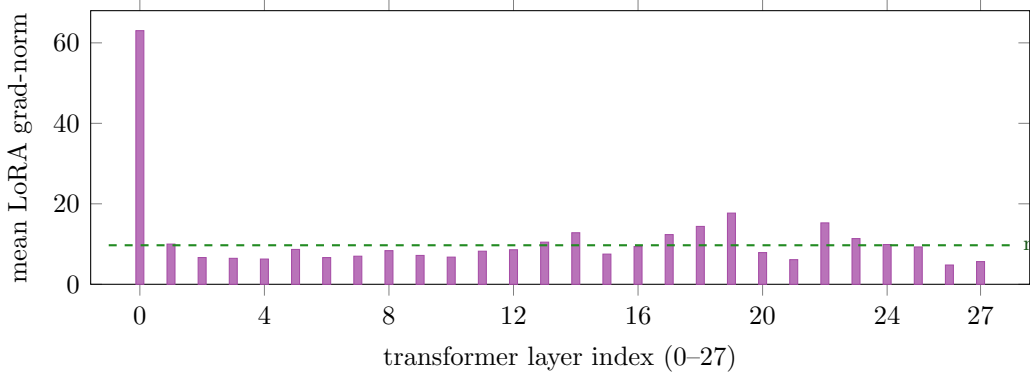


Figure 6.2: P1 white-box (Colab L4, real data). Layer 0 dominates adaptation; a secondary mid-late band (layers 17–23) is active. Top 25% of layers carry 47.6% of gradient (moderate concentration — which *holds* at 3B, 0.39). At this toy scale the step-1 top- k matched the final top- k (overlap = 1.0), but a scaled 3B/four-seed re-test drops the mean to 0.083 ± 0.048 ($\approx$ chance): predictability does *not* survive scaling.

6.6 P1 — Step-1 Predictability Holds at Toy Scale but Collapses at 3B

The white-box layer profile at toy scale (Qwen2.5-1.5B, 28 layers, 5 GRPO-style steps, a hardcoded arithmetic set) initially looked like the clearest positive in the portfolio: the top- k highest-gradient layers after a *single* step were identical to the top- k over all steps (`step1_predicts_final_topk_overlap` = 1.0), suggesting one could decide what to freeze after one update. **This did not survive scaling.** A four-seed re-test (Qwen2.5-3B, 36 layers, 10 steps, *real* GSM8K, seeds {0, 1, 2, 3}) drops the mean step-1→final overlap to 0.083 ± 0.048 — i.e. $\approx$ chance. The mid-late band that dominates by end-of-training *emerges* over steps and is invisible at step 1, so the apparent predictability was an artifact of the toy setup. Adaptation is only *moderately* concentrated: the top 25% of layers carry $\sim 48\%$ of the total gradient norm (`concentration_top25pct_share` = 0.476 at 1.5B, 0.39 at 3B), not the $> 90\%$ that a “freeze almost everything” story would need.

Freeze rule	Cold layers	Freeze-able frac.	Est. LoRA-bwd FLOP saving [†]
Conservative ($g < 0.6 \bar{g}$)	5/28	18%	$\sim 18\%$
Moderate ($g < 0.7 \bar{g}$)	11/28	39%	$\sim 39\%$

Table 6.5: Freeze-able fraction and estimated FLOP saving. [†]**Estimate, not measured.** Assumes per-layer LoRA backward cost is uniform, so freezing k of 28 layers saves $\approx k/28$ of the *LoRA-backward* FLOP only; the frozen base-model forward/backward (which dominates end-to-end training FLOP) is unchanged. Derived from a 1.5B, 5-step proof-of-concept; a scaled run is needed to confirm.

6.7 P8 — Telemetry Separates Method Traces in a Small Proof of Concept

Using the per-step telemetry rather than reward alone, an exploratory classifier separates the four method traces with row-wise cross-validated AUROC 0.838 ± 0.010 , versus 0.426 ± 0.027 for reward-only features. This shows that the stored telemetry contains method-identifying structure beyond the reward summary; it does *not* yet establish run-integrity detection or generalisation. The result is recomputed from the raw `n2_reward_tensor_resume` tensors by `p8_detector.py`: four reward-optimisation algorithms each contribute 40 per-step rows ($N = 160$); the positive “surrogate” class is three GRPO variants reported as GRPO (120 rows) and the negative class is genuine GRPO (40 rows), a 3:1 imbalance. Features are derived entirely from the raw reward/length/advantage arrays (`zvf`, `reward_mean`, `reward_std`, `frac_all_zero`, `frac_all_one`, `mean_len`, `cv_len`, `len_reward_corr`, `adv_abs_mean`, `adv_std`, `lag1_autocorr`). A logistic-regression classifier (median-imputed, standardised, 5-fold stratified CV) reaches AUROC = 0.838 ± 0.010 , versus 0.426 ± 0.027 for a reward-only auditor (below chance). **Overfitting caveat:** with $N = 160$ rows, 11 features, and a 3:1 class imbalance, one trajectory per method, temporal correlation between rows, and row-wise rather than run-wise splits, the AUROC is susceptible to severe optimism. It should be read only as a *method-classification proof of concept*. A larger, multi-run evaluation with grouped temporal/run-level holdout (diverse base models/tasks, thousands of steps) is required before any integrity claim. Remaining provenance caveat: the base model/task are not recorded in the tensor rows, so provenance is incomplete — exactly the gap the P5 protocol is designed to close.

6.8 Cross-Study Synthesis

Read together, the eight studies tell a coherent story rather than eight isolated results. P2 establishes that the dominant measured phenomenon in the fixed study is advantage-signal starvation ($ZVF \approx 0.72\text{--}0.77$), and that it is *easy-driven* (the collapsed groups are mostly already-solved prompts). That immediately rules out the tempting fix — re-baselining — because P2 also shows the “recovered” signal is just a difficulty proxy with no learning content. The natural next lever, a curriculum that removes the waste, is exactly what the curriculum and matched-token studies test; both return honest nulls. P3 tells the same cautionary tale for group size: a single-seed sweep does not establish a universal optimum. P1 extends the pattern to a structural claim — step-1 predictability of layer adaptation — which likewise fails to survive scaling. P8 adds a narrower systems observation: telemetry separates the four available method traces better than reward-only features, but integrity generalisation remains untested. The synthesis, then, is that on this compute scale the *measurement* is useful while most *algorithmic* levers are noise-limited — which is what justifies steering Phase-2 toward the systems/provenance direction (P5) rather than chasing another GRPO variant.

6.9 Rigour and Threats to Validity

The methodology is deliberately adversarial toward its own results.

- **Multi-seed with matched baselines where feasible.** The curriculum intervention is re-tested across three seeds at matched token budget; the P1 scaling check uses four seeds. P2 remains a shared-schedule seed-0 diagnostic, P3 remains a single-seed sweep, and P8 remains one trajectory per method; these are explicitly labelled exploratory rather than confirmatory.
- **Held-out evaluation.** Gains are measured on held-out prompts, and we report the held-out n (8–20) alongside every gain so the reader can see that magnitudes are noise-limited.
- **Recompute-from-tensors.** Published diagnostics are recomputable from stored raw tensors (e.g. `p2_collapse_analysis.py`), not merely read back from a logged summary.
- **Adversarial review plus deterministic verification.** Critical review of the early single-seed conclusions prompted the multi-seed and matched-token re-tests. The evidence retained in the submission is the script-based recomputation: ZVF 0.72–0.77, all-correct 0.65–0.71, and the stored P8 row-CV output.

Known threats we do *not* hide: small held-out sets; a partially-recorded P3 sweep artefact (G8/G16 summaries lost to the W&B thread-safety bug); a deterministic prompt schedule shared across the four P2 methods; and an invalidated `zero_loss_frac` in the first campaign batch (since fixed). Re-running P3 under the process-per-run launcher yields corrected G8/G16 held-out gains of +0.083 (both from $0.750 \rightarrow 0.833$, single seed), superseding the thread-bug-affected values — but these remain single-seed point estimates pending multi-seed replication.

Chapter 7

Conclusions and Future Work

7.1 Conclusions

This phase delivered a working unified GRPO/RLVR benchmark, a per-step diagnostic telemetry layer, and eight studies whose honest verdict is: the algorithmic levers we could test on this setup (naive curriculum, integer group-size sweeps) do *not* produce reliable held-out gains, while the large descriptive effect in the fixed P2 study is *advantage-signal starvation* ($ZVF \approx 0.72\text{--}0.77$). Reporting those negatives — rather than the cherry-picked single-seed positives — is the contribution: it is what makes the portfolio defensible and the benchmark trustworthy. P8 supplies a proof-of-concept systems result: telemetry separates the available method traces better than reward-only features, but cross-run integrity generalisation is untested.

Where the original value is. The strongest *original* bet is **P5, a GRPO provenance / reporting standard**. A targeted search of the cited GRPO, model-reporting, and provenance literature through July 2026 did not identify an equivalent GRPO-specific machine-readable run datasheet with rollout provenance; this is an evidence-bounded novelty claim, not proof of non-existence. The earlier second bet — **P1 predictive layer-freezing** — is now weakened: its step-1 predictability held at 1.5B but collapsed to chance (mean 0.083 ± 0.048) on the scaled 3B four-seed re-test.

7.2 Future Work (Phase-2 Plan)

1. **Token-budget-optimal curriculum.** Replace naive “filter collapsed groups” with an allocation that trades oversampling against training under

an explicit token budget with staleness bounds, tested multi-seed from the start. The Phase-1 null tells us the lever must be *budget-aware*, not just waste-eliminating.

2. **Re-examine P1.** The scaled 3B / multi-seed re-test shows step-1 predictability does *not* hold; the open question is whether the *emerged* mid-late band supports any FLOP-saving freeze schedule when read off late-training gradients rather than step 1.
3. **The P5 protocol.** Specify and implement the GRPO post-training datasheet plus cryptographic rollout provenance, and dogfood it on the harness’s own runs so every benchmark result ships with a verifiable provenance record.
4. **Adequately powered evaluation.** The most important methodological upgrade for Phase-2 is statistical power: move from held-out $n = 8\text{--}20$ to the full GSM8K test split (or a large stratified subset), run ≥ 5 seeds, and report confidence intervals on every headline metric, so that a genuine effect can be distinguished from the noise floor that dominates the Phase-1 nulls.
5. **Cross-framework divergence study.** Execute the identical task/reward/decoding/seed configuration on TRL, veRL, OpenRLHF, and Tinker and publish the divergence table — the concrete demonstration of the attribution claim that Phase-1 designs but does not yet run.
6. **Harden the P8 classifier.** Re-train and evaluate the method-trace classifier on a much larger, multi-run, multi-model, multi-task corpus with grouped run-level and temporal held-out splits (not row-CV only) to establish whether the AUROC 0.84 proof-of-concept generalises.
7. **Publication.** Consolidate the diagnostic findings and the P5 standard into a paper draft targeting a reputed peer-reviewed venue, as required by the programme guidelines.

References

- [1] Z. Shao, P. Wang, Q. Zhu, R. Xu, J. Song, et al. DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models. *arXiv:2402.03300*, 2024. (Introduces GRPO.)
- [2] D. Guo, D. Yang, H. Zhang, et al. (DeepSeek-AI). DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. *arXiv:2501.12948*, 2025.
- [3] A. Ahmadian, C. Cremer, M. Gallé, et al. Back to Basics: Revisiting REINFORCE-Style Optimization for Learning from Human Feedback in LLMs (RLOO). *arXiv:2402.14740*, 2024.
- [4] R. Rafailov, A. Sharma, E. Mitchell, S. Ermon, C. D. Manning, C. Finn. Direct Preference Optimization: Your Language Model is Secretly a Reward Model. *NeurIPS*, 2023.
- [5] Z. Liu, C. Chen, W. Li, et al. Understanding R1-Zero-Like Training: A Critical Perspective (Dr. GRPO). *arXiv:2503.20783*, 2025.
- [6] Q. Yu, Z. Zhang, R. Zhu, et al. DAPO: An Open-Source LLM Reinforcement Learning System at Scale. *arXiv:2503.14476*, 2025. (Dynamic sampling to avoid zero-variance groups.)
- [7] L. Ouyang, J. Wu, X. Jiang, et al. Training Language Models to Follow Instructions with Human Feedback. *NeurIPS*, 2022.
- [8] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, O. Klimov. Proximal Policy Optimization Algorithms. *arXiv:1707.06347*, 2017.
- [9] J. Schulman, P. Moritz, S. Levine, M. Jordan, P. Abbeel. High-Dimensional Continuous Control Using Generalized Advantage Estimation. *arXiv:1506.02438*, 2015.
- [10] N. Lambert, J. Morrison, V. Pyatkin, et al. TüLU 3: Pushing Frontiers in Open Language Model Post-Training (RLVR). *arXiv:2411.15124*, 2024.
- [11] G. Sheng, C. Zhang, Z. Ye, et al. HybridFlow: A Flexible and Efficient RLHF Framework (veRL). *arXiv:2409.19256*, 2024.

- [12] J. Hu, X. Wu, W. Wang, et al. OpenRLHF: An Easy-to-Use, Scalable and High-Performance RLHF Framework. *arXiv:2405.11143*, 2024.
- [13] L. von Werra, Y. Belkada, L. Tunstall, et al. TRL: Transformer Reinforcement Learning. <https://github.com/huggingface/trl>, 2020–.
- [14] K. Cobbe, V. Kosaraju, M. Bavarian, et al. Training Verifiers to Solve Math Word Problems (GSM8K). *arXiv:2110.14168*, 2021.
- [15] M. Chen, J. Tworek, H. Jun, et al. Evaluating Large Language Models Trained on Code (HumanEval). *arXiv:2107.03374*, 2021.
- [16] T. Gebru, J. Morgenstern, B. Vecchione, et al. Datasheets for Datasets. *Communications of the ACM*, 64(12):86–92, 2021.
- [17] M. Mitchell, S. Wu, A. Zaldivar, et al. Model Cards for Model Reporting. *ACM FAT**, 2019.
- [18] Z. Zhang, Z. Han, C. Mavromatis, et al. Train Less, Learn More: Adaptive Efficient Rollout Optimization (AERO) for Group-Based Reinforcement Learning. *arXiv:2602.14338*, 2026.
- [19] G. Nan, S. Chen, J. Huang, et al. NGRPO: Negative-enhanced Group Relative Policy Optimization. *arXiv:2509.18851*, 2025.
- [20] K. Zhang, Y. Hong, J. Bao, et al. GVPO: Group Variance Policy Optimization for Large Language Model Post-Training. *arXiv:2504.19599*, 2025. NeurIPS 2025.
- [21] Y. Wang, J. Zhao, C. Zhao, et al. λ -GRPO: Unifying the GRPO Frameworks with Learnable Token Preferences. *arXiv:2510.06870*, 2025.
- [22] Y. Kim. MC-GRPO: Median-Centered Group Relative Policy Optimization for Small-Rollout Reinforcement Learning. *arXiv:2601.22582*, 2026.
- [23] X. He, Q. Sun, A. Cheng, et al. Advantage Collapse in Group Relative Policy Optimization: Diagnosis and Mitigation (AVSPO). *arXiv:2605.21125*, 2026. ICML 2026.

Appendix A

Complete Run Registry

Table A.1 lists the runs behind the Phase-1 findings. All runs log to Weights & Biases (project `rlvr-openings`) with per-step telemetry and saved group reward tensors; runs are pinned by config, seed, and framework.

Study	Model / setup	Seeds	Primary output
P1 (toy)	Qwen2.5-1.5B, 28 layers, 5 steps, arithmetic	1	per-layer grad-norm; overlap= 1.0
P1 (scaled)	Qwen2.5-3B, 36 layers, 10 steps, GSM8K	{0, 1, 2, 3}	overlap 0.083 ± 0.048 ; top-25% share 0.39
P2	4 methods, 40 steps, 16×8 , GSM8K	shared seed-0 sched.	ZVF 0.72–0.77
P3	Qwen3.5-4B, $G \in \{2, 4, 8, 16\}$, 6 steps	0	token cost 9.7k–68k; exploratory sweep
Curriculum	Qwen3.5-4B, 8 steps, mixed-variance filter	0	ZVF $0.50 \rightarrow 0.00$, $4.81 \times$ cost
Matched-token	Qwen3.5-4B, $G=4$, baseline vs curriculum, ~ 30 k tokens	{0, 1, 2}	both arms +0.028 mean
P8	method-trace classifier, $N = 160$ correlated rows, 5-fold row CV	1 trace/method	AUROC 0.838 ± 0.010

Table A.1: Complete Phase-1 run registry.

Appendix B

Reproducibility and Provenance

Every published diagnostic in this report is regenerable from stored artefacts:

- `p2_collapse_analysis.py` regenerates the ZVF / collapse JSON from the saved P2 reward tensors (reproduces $ZVF = 0.72\text{--}0.77$).
- `p8_detector.py` regenerates `detector_reproduced.json` from the `n2_reward_tensor_resume` tensors (reproduces row-CV AUROC 0.838).
- `experiments/results/token_budget/results.json` records the matched-token three-seed baseline-vs-curriculum comparison.

Platforms and budget: managed GRPO training via Tinker; white-box P1 on a Google Colab L4 GPU; batch scoring on Modal; logging on Weights & Biases. The full Phase-1 empirical portfolio was executed within a modest budget (order of tens of US dollars of managed compute plus free-tier GPU time).

Adversarial review and verification. AI-assisted adversarial review was used during development to identify overclaims and prompt additional checks. These model assessments are not treated as scientific evidence and are not required to reproduce any number in the report. Verification in the submission is script-based, using stored tensors and explicit result JSONs as described above.

AI-assistance disclosure. Language models assisted with adversarial review, code review, and editorial revision. The candidate remains responsible for the research design, execution choices, source verification, calculations, and final claims. No model-generated verdict is reported as an empirical result.

Code availability. All harness code, diagnostic scripts, raw reward tensors, and result JSONs are maintained in the project repository under version control,

pinned by configuration, seed, and framework; the run registry (Table A.1) keys each finding to its producing run. The complete report source, figures, and this document are likewise version-controlled.

Originality / plagiarism. This report is an original individual work. All external sources and prior art are cited (Chapter 2, References). The report has been prepared for the institutional similarity (plagiarism) check with a target similarity of $\leq 15\%$ in line with the programme guidelines, and reuses the candidate’s own prior project write-ups where applicable.

Appendix C

Notation, Symbols, and Abbreviations

Term / symbol	Meaning
GRPO	Group Relative Policy Optimization; the group-baseline RL post-training algorithm studied here [1].
RLVR	Reinforcement Learning with Verifiable Rewards; a programmatic checker supplies the reward with no learned reward model.
ZVF (ZVF)	Zero-Variance Fraction; the fraction of prompt-groups in a step whose within-group reward variance is zero (Eq. (4.2)).
G	Group size; number of sampled completions drawn per prompt.
A_i	Group-relative advantage of completion i (Eq. (4.1)).
μ_g, σ_g	Within-group reward mean and standard deviation.
β	KL-penalty coefficient toward the frozen reference policy.
Gradient utilisation	Complementary fraction of groups with non-zero reward-relative advantage signal ($1 - \text{ZVF}$ at the group level); excludes KL-only updates.
Held-out gain	Change in accuracy on a held-out prompt set; always reported with its denominator n .
All-correct / all-wrong	Decomposition of collapsed (zero-variance) groups by whether every completion was right or wrong.
Matched baseline	A control run identical to the treatment except for the single lever under study.
AUROC	Area under the ROC curve; the exploratory P8 method-trace classifier metric.